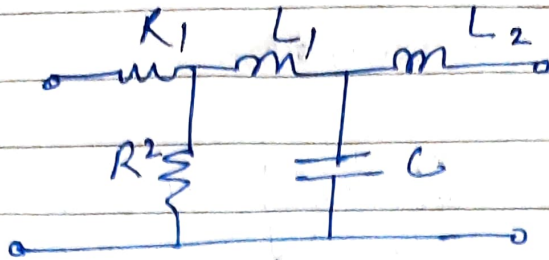


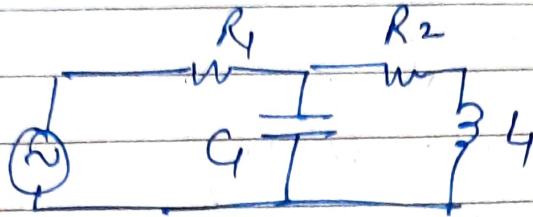
Circuit and System

1. Network and Circuit :-

Network :-



Circuit :-



2. Circuit Elements (R, L, C, gen^r) may be :-

- (a) Active or Passive
- (b) Unilateral or Bilateral :- eg :-
 - ↳ diode, transistor
 - ↳ t/m line

(c) Lumped or distributed :-

(d) Linear or Non-linear :-

Principle of Homogeneity :-

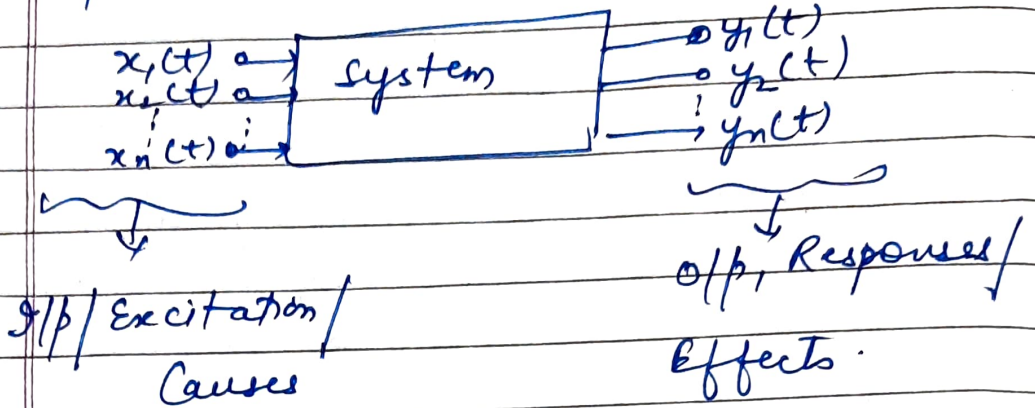
$$\begin{aligned} x &\rightarrow [N/W] \rightarrow y \\ 2x &\rightarrow [N/W] \rightarrow 2y \\ \vdots & \\ nx &\rightarrow [N/W] \rightarrow ny \end{aligned}$$

Principle of Superposition :-

$$\begin{aligned} x_1 &\rightarrow [N/W] \rightarrow y_1 & x_2 &\rightarrow [N/W] \rightarrow y_2 \\ \therefore x_1 + x_2 &\rightarrow [N/W] \rightarrow y_1 + y_2 \end{aligned}$$

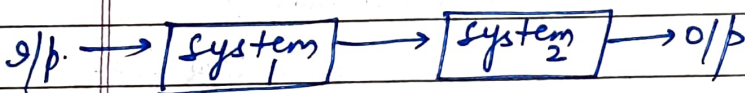
SYSTEM:-

Representation 1:-

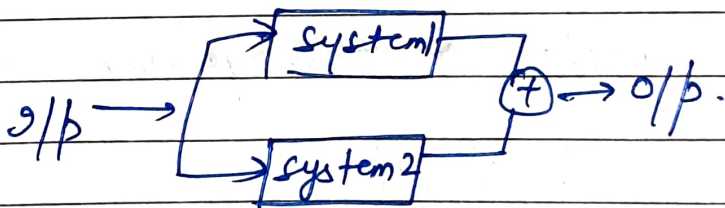


Interconnection of System:-

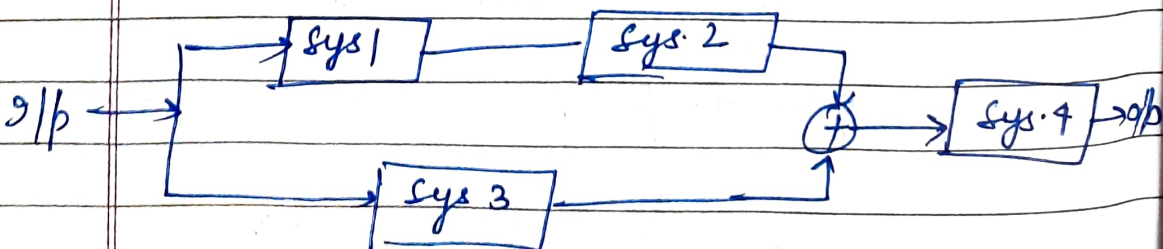
(a) Series (Cascade interconnection)



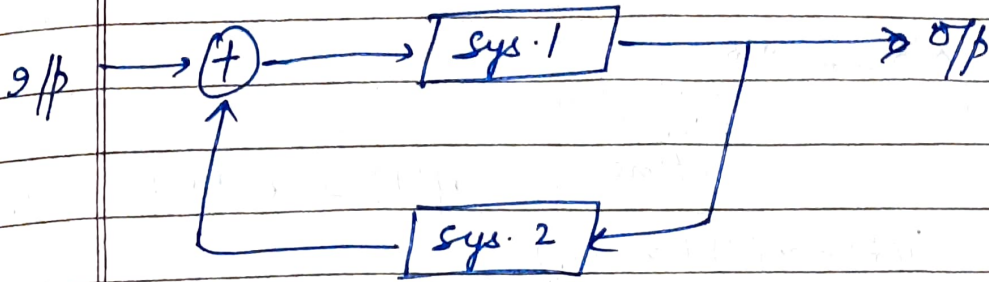
(b) parallel Interconnection:-



(c) Series-// Interconnection:-

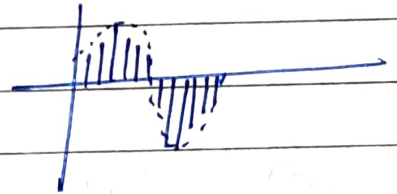
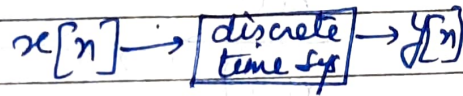
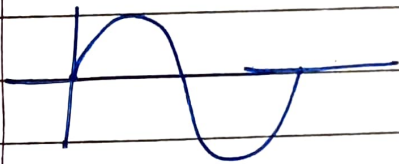
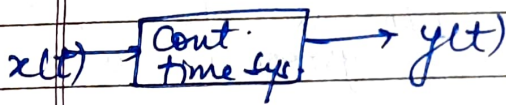


② Feedback Interconnection :-



Basic System properties :-

① Continuous time and Discrete time systems



② Time Variant and Time-invariant System

③ Linear and Non-linear System :-

④ Instantaneous and dynamic system :-

$$o/p = f(i/p \text{ present time}) \quad ; \quad o/p = f(i/p \text{ past or future values})$$

$$y(t) = 3x(t) + 2x(t)$$

$$y[n] = 3x[n]$$

or

$$o/p = f(\text{past } o/p + \text{present } i/p)$$

$$y(t) = 3x(t) - x^2(t)$$

$$y[n] = 3x[n] - x^2[n]$$

(5) Causal and Non-Causal System :-

O/p = f (I/p_{present} & past time)

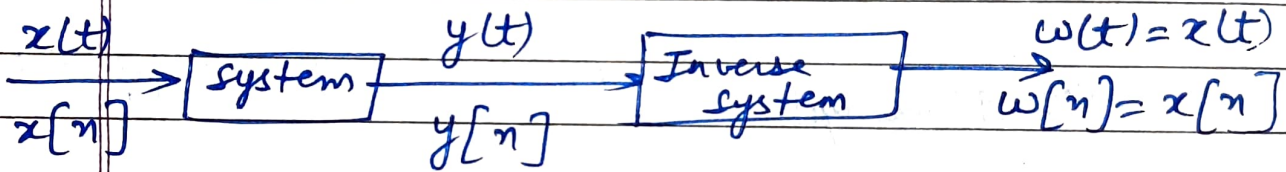
$$y(t) = x(t) + x(t-1)$$

$$y[n] = x[n] + x[n-1]$$

$$y(t) = x(t) + x(t+1)$$

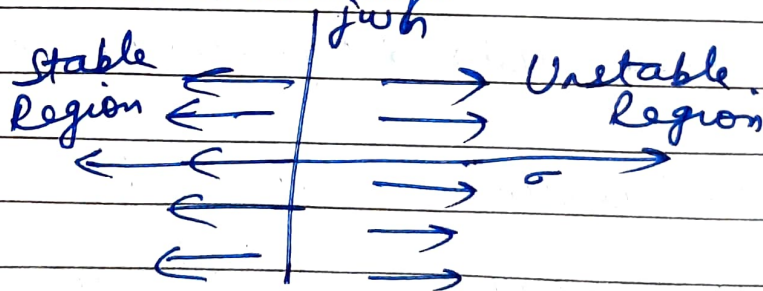
$$y[n] = x[n] + x[n+1]$$

(6) Invertibility and Inverse System :-

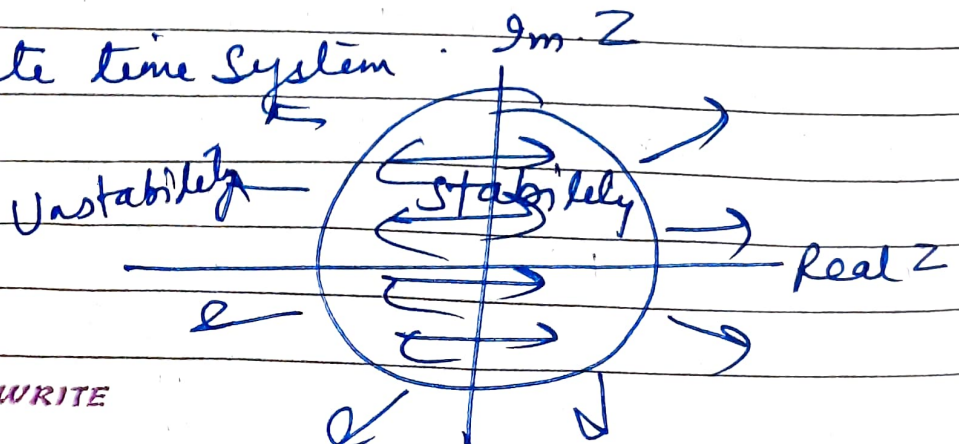


(7) Stability :-

for continuous Time System :-



Discrete time System



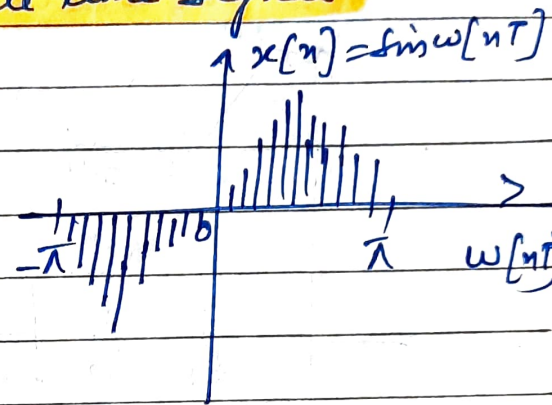
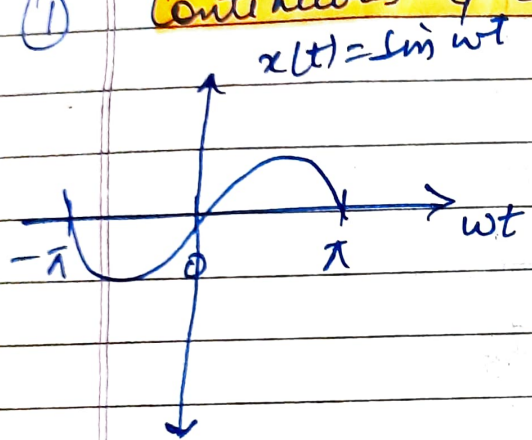
Signal & waveform synthesis :-

Signal :- / functions :-

I_s and V_s are fun. of time.
These funs of time are called signals.

Classification of signals :-

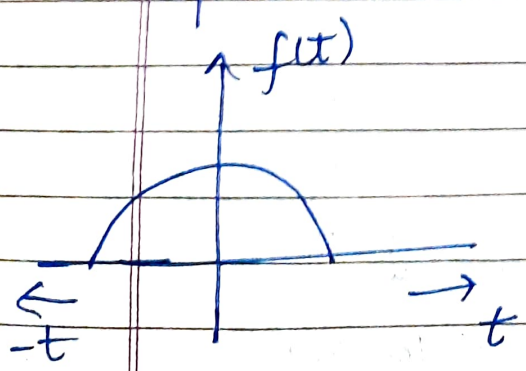
① Continuous & discrete time signal :-



② Even and odd signal :-

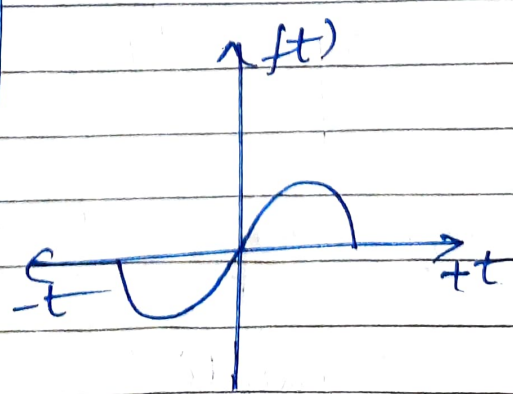
$$f(-t) = f(t)$$

Signal is symmetrical across the vertical axis in its domain of representation



$$f(-t) = -f(t)$$

It is -ve of its reflection



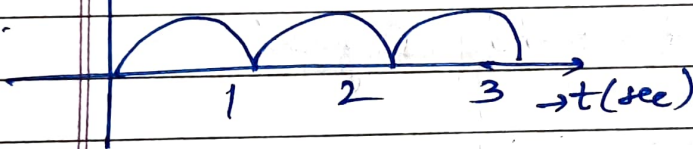
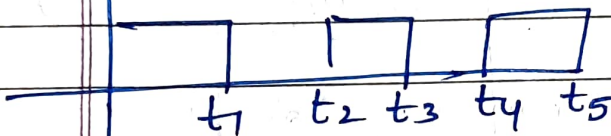
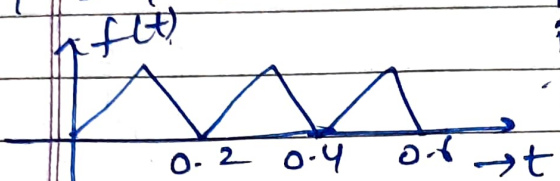
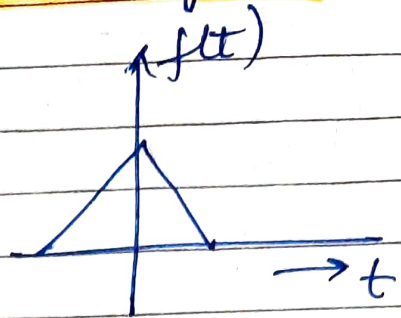
③

Periodic and Aperiodic Signals :-

$$f(t) = f(t+T)$$

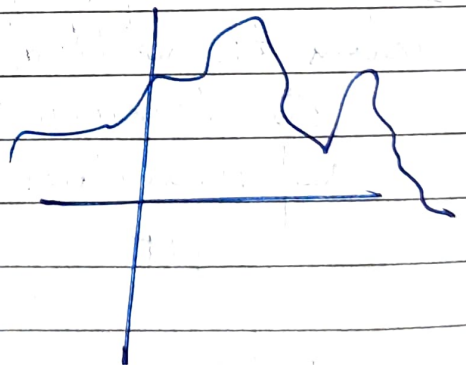
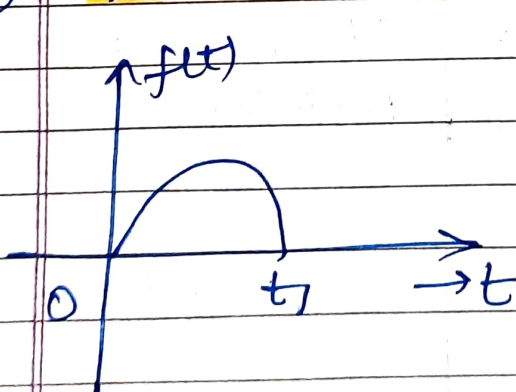
$$[-\infty < t < \infty]$$

where $x(t)$ has a
fundamental period T
 $T = +ve \text{ constt.}$



④

Deterministic and Random Signals :-



⑤

Power and Energy Signals :-

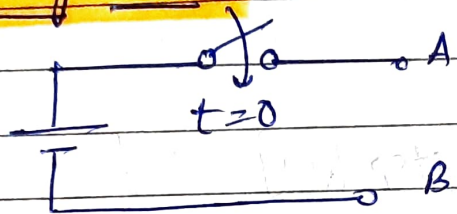
Periodic &
Random Signals

Deterministic
and Aperiodic Signals

STANDARD SIGNALS OR SINGULARITY FNCT.

- | | |
|---------------|-----------------------|
| ① Step signal | ① Unit doublet signal |
| ② Ramp " | ② Exponential signal |
| ③ Impulse " | ③ Sinusoidal " |
| | ④ Gate " |

① Step signal :-

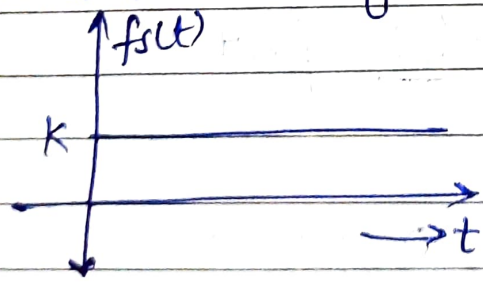


Step fn / signal $f_s(t)$ is defined by

$$f_s(t) = \begin{cases} 0; & t < 0 \\ k; & t > 0 \end{cases}$$

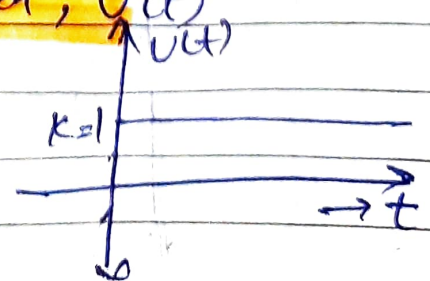
k = Amplitude of the step signal

Note that the fn is undefined at $t = 0$



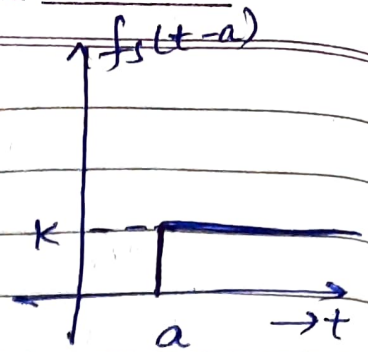
If the value of $k=1$, then Step signal $f_s(t)$ is called unit step fn, $U(t)$

$$U(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$

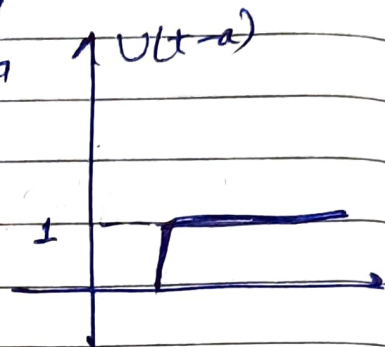


shifted / delayed fcn -

shifted step fcn $\left\{ \begin{aligned} f_s(t-a) &= \begin{cases} 0 & ; t < a \\ k & ; t > a \end{cases} \end{aligned} \right.$ and



shifted unit step fcn. $\left\{ \begin{aligned} U(t-a) &= \begin{cases} 0 & ; t < a \\ 1 & ; t > a \end{cases} \end{aligned} \right.$



* $L[\text{unit step fcn}] = \frac{1}{s}$

* $L[\text{delayed unit step}] = \frac{e^{-as}}{s}$

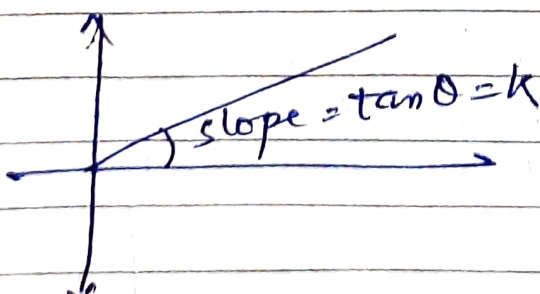
* $L[kU(t)] = \frac{k}{s}$

* $L[kU(t-a)] = \frac{ke^{-as}}{s}$

② Ramp Signal - $f_r(t)$

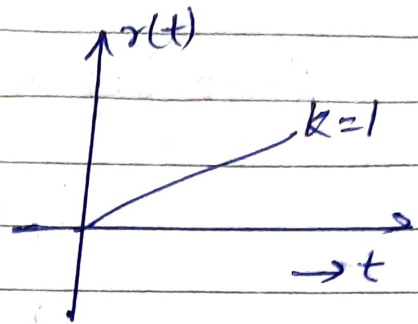
$$f_r(t) = \begin{cases} 0 & ; t < 0 \\ kt & ; t \geq 0 \end{cases}$$

$k = \text{slope of ramp signal}$



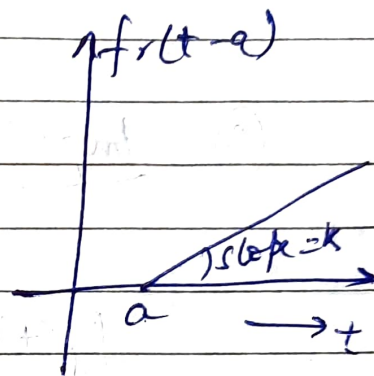
Unit ramp signal :- of slope, $k=1$. It is denoted by $r(t)$

$$r(t) = \begin{cases} 0, & t < 0 \\ t, & t \geq 0 \end{cases}$$



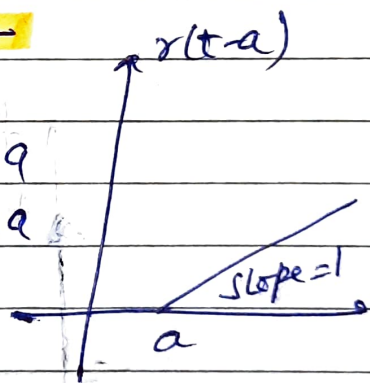
Delayed/Shifted ramp signal :-

$$f_r(t-a) = \begin{cases} 0; & t < a \\ k(t-a); & t \geq a \end{cases}$$



Shifted unit ramp signal :-

$$r(t-a) = \begin{cases} 0 & ; t < a \\ (t-a) & ; t \geq a \end{cases}$$



* $L[\text{unit ramp } f_{xu}] = \frac{1}{s^2}$

* $L[\text{Ramp}] = \frac{k}{s^2}$

* $L[\text{delayed Ramp}] = \frac{ke^{-as}}{s^2}$

* $L[\text{delayed unit ramp}] = \frac{e^{-as}}{s^2}$

* Differentiation of ramp f_{xu} gives step f_{xu} .

Unit ramp signal :-

$$r(t) = tU(t)$$

$$r(t-a) = (t-a)U(t-a)$$

Ramp signal :-

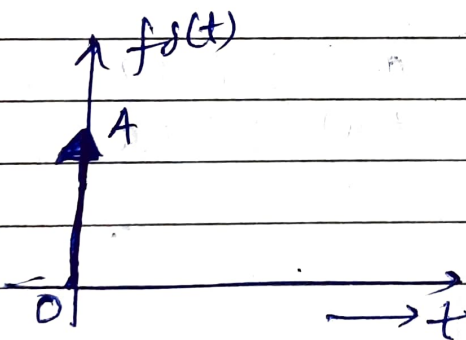
$$fr(t) = kr(t) = k \cdot U(t)$$

$$fr(t-a) = kr(t-a) = k(t-a)U(t-a)$$

③ Impulse signal / Dirac delta signal :- $f_s(t)$

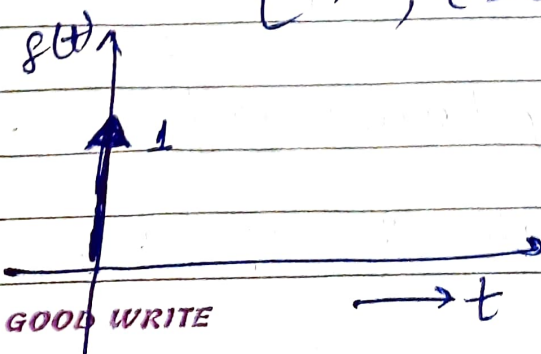
$$f_s(t) = \begin{cases} 0, & t \neq 0 \\ A, & t = 0 \end{cases}$$

A - Amplitude of impulse signal.



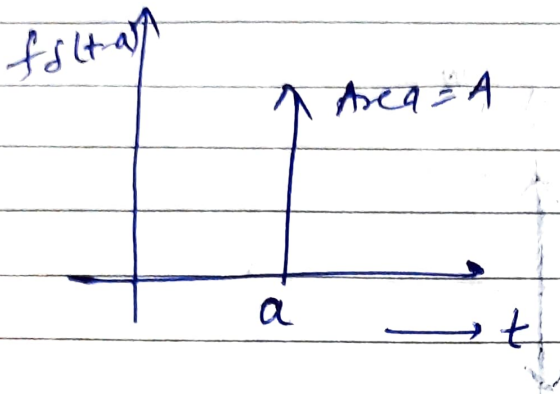
Unit Impulse signal :- $\delta(t)$

$$\delta(t) = \begin{cases} 0, & t \neq 0 \\ 1, & t = 0 \end{cases}$$



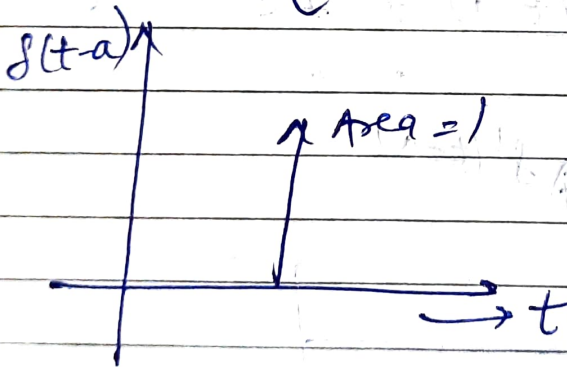
Shifted Impulse Signal :-

$$f_s(t-a) = \begin{cases} 0, & t \neq a \\ A, & t = a \end{cases}$$



Shifted unit Impulse signal :-

$$s(t-a) = \begin{cases} 0, & t \neq a \\ 1, & t = a \end{cases}$$



* $\mathcal{L}[\text{Impulse fn}] = 1$

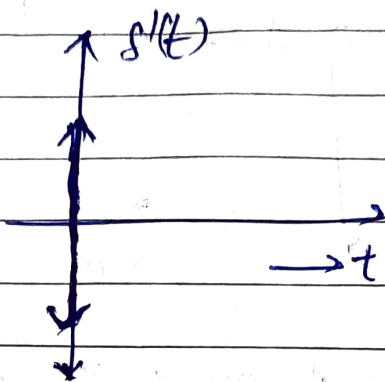
* Derivative of Step fn = Impulse fn.

OTHER BASIC SIGNALS:-

① Unit doublet signal:-

$$s'(t) = \frac{d}{dt} [s(t)] = +\infty \text{ \& } -\infty ; t=0$$

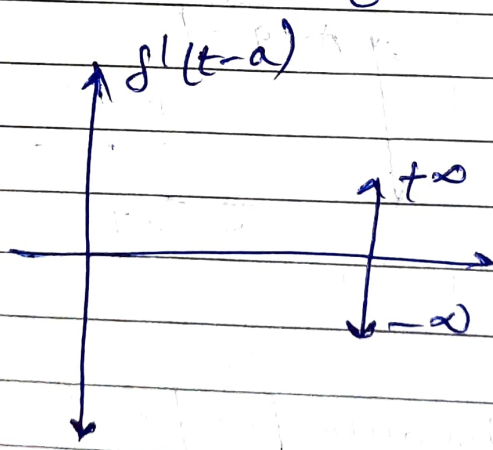
$$= 0 ; t \neq 0$$



Shifted unit doublet signal:-

$$s'(t-a) = +\infty \text{ \& } -\infty ; t=a$$

$$= 0 ; t \neq a$$

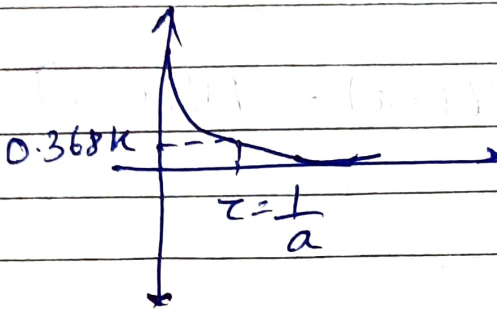


* Derivative of unit Impulse signal =
unit doublet signal

② Exponential signal :-

$$f(t) = \begin{cases} 0 & ; t < 0 \\ ke^{-at} & ; t \geq 0 \end{cases}$$

a & k are real const^s

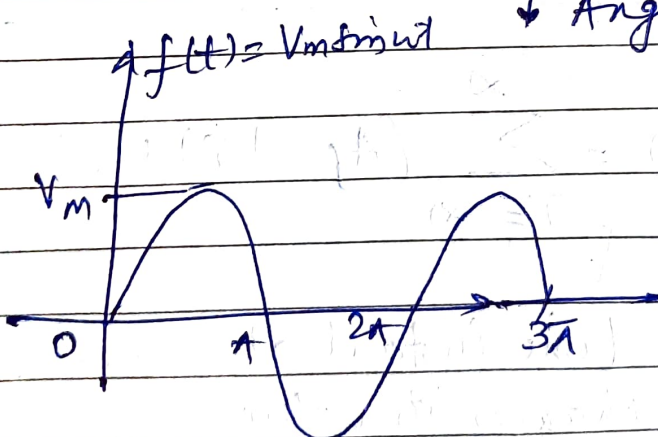


③ Sinusoidal signal :-

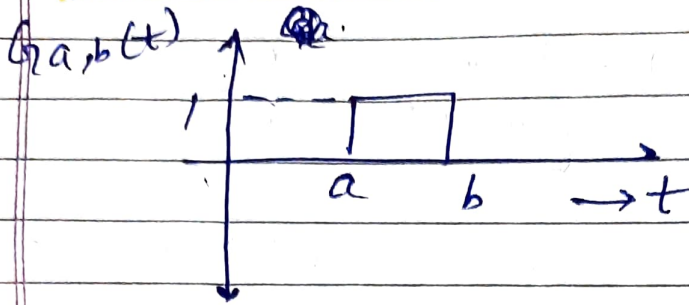
$$f(t) = \begin{cases} 0 & ; t < 0 \\ V_m \sin \omega t & ; t \geq 0 \end{cases}$$

↓ Peak Amplitude

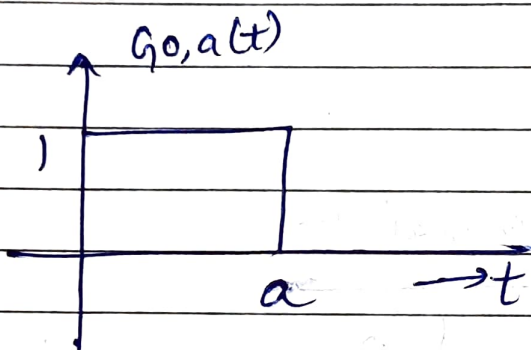
↓ Angular frequency



④ Gate signal:-



$$G_{a,b}(t) = U(t-a) - U(t-b)$$



$$G_{0,a}(t) = U(t) - U(t-a)$$

⑤ Direct formula (k.M formula):-

$$f(t) = \sum_{T=-\infty}^{\infty} (A_f - A_i) U(t-T)$$

T = time instant at which the $f(t)$ changes its value

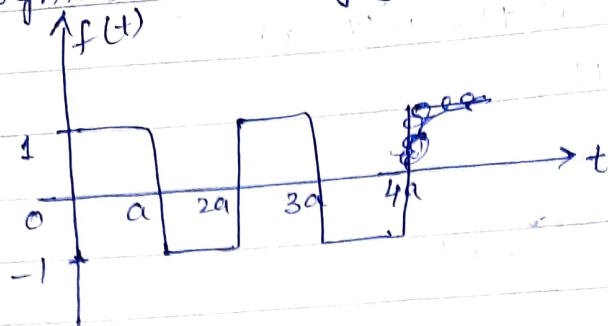
A_f & A_i = final and initial values at the corresponding time instt.

If $f(t)$ exist only for $t \geq 0$,

GOOD WRITE

$$f(t) = \sum_{T=0}^{\infty} (A_f - A_i) U(t-T)$$

e.g 1) - Synthesize using gate fun? -



Soln:- Using gate fun? -

$$f(t) = (+1) G_{0,a}(t) + (-1) G_{a,2a}(t) + (+1) G_{2a,3a}(t) + (-1) G_{3a,4a}(t) + \dots$$

$$f(t) = 1 [U(t-0) - U(t-a)] + (-1) [U(t-a) - U(t-2a)] + (+1) [U(t-2a) - U(t-3a)] + (-1) [U(t-3a) - U(t-4a)] + \dots$$

$$= U(t) - U(t-a) - U(t-a) + U(t-2a) + U(t-2a) - U(t-3a) - U(t-3a) + U(t-4a) + \dots$$

$$= U(t) - 2U(t-a) + 2U(t-2a) - 2U(t-3a) + \dots \quad \underline{\underline{\text{Ans}}}$$

Using direct formula: -

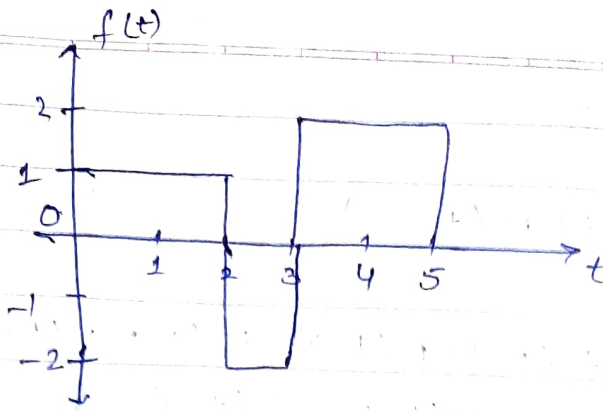
$$f(t) = \sum_{T=0}^{\infty} (A_f - A_i) U(t-T)$$

where T = is the time at which fcn. changes its values.

$$= (1-0)U(t-0) + (-1-1)U(t-a) + [1-(-1)]U(t-2a) + (-1-1)U(t-3a) + \dots$$

$$= U(t) - 2U(t-a) + 2U(t-2a) - 2U(t-3a) + \dots \quad \underline{\underline{\text{Ans}}}$$

eg2:-



Soln:- Gate fun:-

$$f(t) = (+1) G_{0,2}(t) + (-2) G_{2,3}(t) + (+2) G_{3,5}(t)$$

$$f(t) = (1) [U(t-0) - U(t-2)] + (-2) [U(t-2) - U(t-3)] + (+2) [U(t-3) - U(t-5)]$$

$$= U(t) - 3U(t-2) + 4U(t-3) - 2U(t-5) \quad \underline{\underline{\text{Ans}}}$$

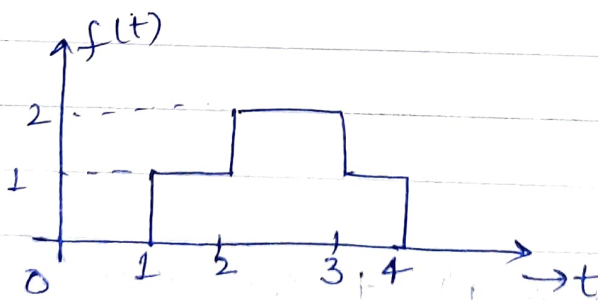
K.M formula:-

$$f(t) = \sum_{T=0}^{\infty} (A_f - A_i) U(t-T)$$

$$= (1-0) U(t-0) + [-2-(1)] U(t-2) + [2-(-2)] U(t-3) + (0-2) U(t-5)$$

$$= U(t) - 3U(t-2) + 4U(t-3) - 2U(t-5) \quad \underline{\underline{\text{Ans}}}$$

eg3:-



Soln:- Gate fun:-

$$f(t) = (+1) G_{0,1}(t) + (1) G_{1,2}(t) + (2) G_{2,3}(t) + (1) G_{3,4}(t)$$

$$= (+1) [U(t-0) - U(t-1)] + (+1) [U(t-1) - U(t-2)] + (+2) [U(t-2) - U(t-3)] + (+1) [U(t-3) - U(t-4)]$$

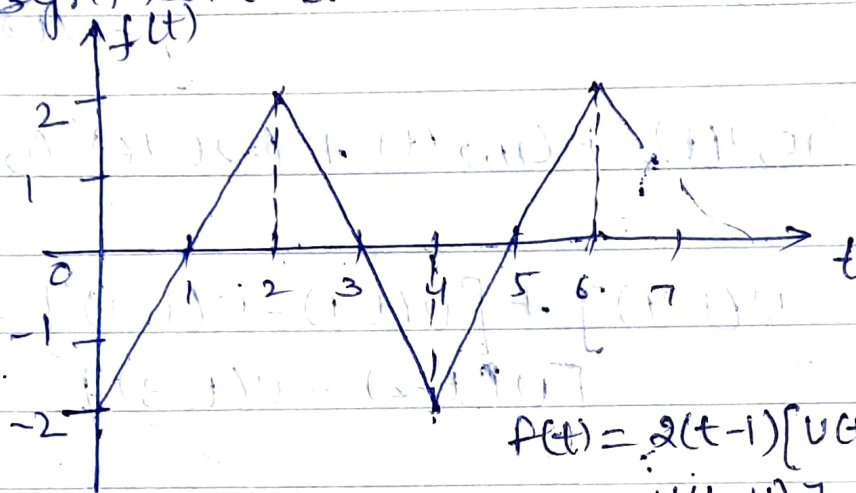
$$= U(t-0) + U(t-1) - U(t-2) - U(t-3) - U(t-4) \quad \underline{\underline{\text{Ans}}}$$

Using k-M formula -

$$f(t) = \sum_{T=0}^{\infty} (A_f - A_i) U(t-T)$$

$$= (1-0) U(t-1) + (2-1) U(t-2) + (1-2) U(t-3) + (0-1) U(t-4)$$
$$= U(t-1) + U(t-2) - U(t-3) - U(t-4) \quad \underline{\underline{\text{Ans}}}$$

eg 5:- Synthesize it :-



slope = $\frac{4}{2}$

$$f(t) = 2(t-1)[u(t-0) - u(t-2)] + (-2)(t-3)[u(t-2) - u(t-4)] + 2(t-5)[u(t-4) - u(t-6)] + \dots$$

Soln:- Using Gate fun :-

$$f(t) = (2t-2)[u(t-0) - u(t-2)] + (-2t+6)[u(t-2) - u(t-4)] + (2t-10)[u(t-4) - u(t-6)] + \dots$$

$$= (2t-2)u(t) + (-4t+8)u(t-2) + (4t-16)u(t-4) + \dots$$

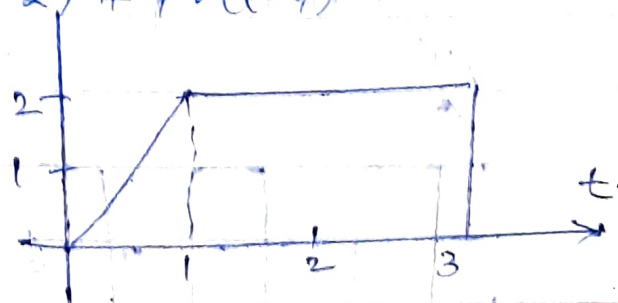
$$= 2(t-1)u(t) - 4(t-2)u(t-2) + 4(t-4)u(t-4) + \dots \quad \underline{\underline{Ans}}$$

$$= 2\delta(t-1)u(t) - 4\delta(t-2) + 4\delta(t-4) + \dots$$

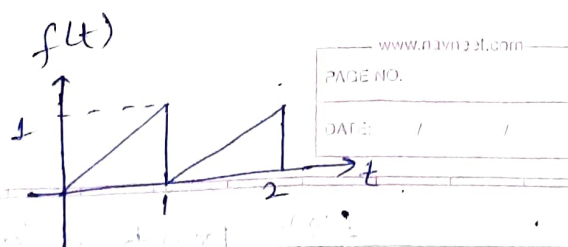
eg 5:- Express using std signal

Soln:- Gate fun:-

$$g(t) = 2(t-0)[u(t-0) - u(t-1)] + 2[u(t-1) - u(t-3)] \quad \underline{\underline{Ans}}$$



slope 1 = $\frac{1}{1}(t-0) = t$
 slope 2 = $\frac{1}{1}(t-1) = (t-1)$



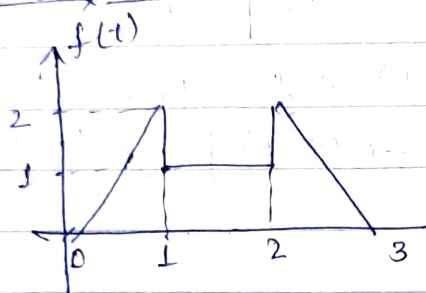
eg 61- In terms of delayed fan?

Soln:-

Gate fan 1-

$$\begin{aligned} f(t) &= t \cdot G_{0,1}(t) + (t-1) G_{1,2}(t) \\ &= t [U(t-0) - U(t-1)] + (t-1) [U(t-1) - U(t-2)] \\ &= t U(t) - U(t-1) - (t-1) U(t-2) \\ &= t U(t) - U(t-1) - (t-2) U(t-2) - U(t-2) \\ &= t U(t) - U(t-1) - t U(t-2) - U(t-2) \quad \underline{\text{Ans}} \end{aligned}$$

eg 71- Using step & ramp fan?



Soln:-

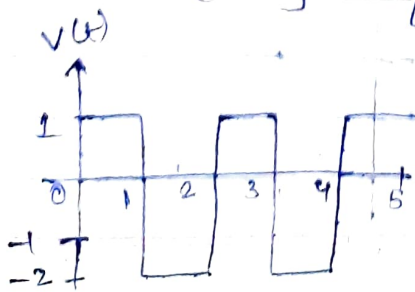
$$\begin{aligned} f(t) &= 2(t-0) G_{0,1}(t) + G_{1,2}(t) + (-2t+6) G_{2,3}(t) \\ &= 2t [U(t-0) - U(t-1)] + [U(t-1) - U(t-2)] + (-2t+6) [U(t-2) - U(t-3)] \\ &= 2t U(t) - (2t-1) U(t-1) - (2t-5) U(t-2) + (2t-6) U(t-3) \quad \underline{\text{Ans}} \end{aligned}$$

eg 81- Sketch the waveform

$$v(t) = u(t) + \sum_{k=1}^{\infty} (-1)^k 3U(t-k)$$

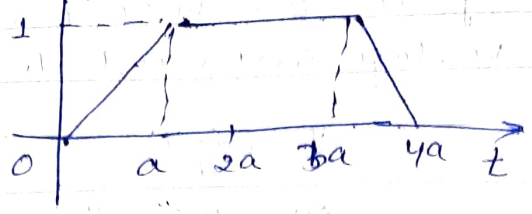
Soln:- Expanding :-

$$\begin{aligned} v(t) &= U(t) - 3U(t-1) + 3U(t-2) - 3U(t-3) + 3U(t-4) + \dots \\ &= [U(t) - U(t-1)] - 2[U(t-1) - U(t-2)] + [U(t-2) - U(t-3)] - 2[U(t-3) - U(t-4)] + \dots \end{aligned}$$



eg 9.1

Using ramp function



Soln:

$$\begin{aligned}
 f(t) &= \frac{1}{a} \cdot t [u(t-0) - u(t-a)] + [u(t-a) - u(t-3a)] \\
 &\quad + \left[\left(-\frac{1}{a} \right) (t-3a) \right] [u(t-3a) - u(t-4a)] \\
 &= \frac{1}{a} t u(t) - \frac{1}{a} (t-a) u(t-a) - \frac{1}{a} (t-3a) u(t-3a) + \\
 &\quad \frac{1}{a} (t-4a) u(t-4a) \\
 &= \frac{1}{a} r(t) - \frac{1}{a} r(t-a) - \frac{1}{a} r(t-3a) + \frac{1}{a} r(t-4a) \quad \underline{\underline{\text{Ans}}}
 \end{aligned}$$

LTI system :- linear time invariant system

Prop:-

- ① Causality
- ② Invertibility
- ③ Stability
- ④ Commutative proper $\rightarrow y = x \cdot h = h \cdot x$
- ⑤ Distributive $\rightarrow y = x (h_1 + h_2) = x \cdot h_1 + x \cdot h_2$
- ⑥ Associative $\rightarrow y = (x \cdot h_1) \cdot h_2 = x \cdot h_1 \cdot h_2$